

MINI SOC PROJECT

Using Elastic Stack (ELK)

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CHAPTER 1

INTRODUCTION

Security Operations Centers (SOCs) are responsible for continuously monitoring, detecting, investigating, and responding to cybersecurity threats. Organizations rely on Security Information and Event Management (SIEM) platforms to collect logs from multiple systems, analyze security events, and generate alerts for suspicious activities.

This project demonstrates the implementation of a Mini SOC using the Elastic Stack (ELK), which consists of Elasticsearch, Kibana, and Elastic Agent. A custom application log source was developed to simulate login events and other operational activities. The logs were collected using Elastic Agent, parsed with an ingest pipeline using Grok patterns, indexed into Elasticsearch, and visualized through Kibana dashboards.

Detection rules were created to identify suspicious activities such as failed login attempts, multiple failed logins, firmware uploads, and network scanning events. The generated alerts were analyzed using the Elastic Security module to demonstrate a simplified SOC workflow.

The objective of this project is to showcase practical knowledge of SIEM deployment, log management, detection engineering, and security monitoring using the Elastic Stack in a controlled lab environment.

CHAPTER 2

PROJECT OBJECTIVES

The objective of this Mini SOC project is to design and implement a Security Operations Center (SOC) environment using the Elastic Stack (ELK). The project demonstrates the complete workflow of centralized log collection, parsing, visualization, and security monitoring in a controlled lab environment.

The primary objectives of this project are:

- Build a centralized log management solution using Elasticsearch, Kibana, and Elastic Agent.
- Collect custom application logs generated from a simulated operational environment.
- Parse unstructured log data using an ingest pipeline with Grok patterns to extract meaningful fields.
- Store the processed logs in Elasticsearch for efficient searching and analysis.
- Create Kibana dashboards to visualize login activity, failed login attempts, operator activity, and other security-related events.
- Develop detection rules to identify suspicious activities such as multiple failed logins, firmware uploads, and network scanning attempts.
- Generate security alerts through the Elastic Security module and demonstrate a simplified SOC monitoring workflow.
- Gain practical experience with SIEM technologies, log analysis, and security monitoring using the Elastic Stack.

CHAPTER 3

PROJECT ARCHITECTURE

The Mini SOC project is built using the Elastic Stack (ELK), consisting of Elasticsearch, Kibana, and Elastic Agent. A custom application running on the Ubuntu server generates operational logs that simulate real-world security events such as login attempts, firmware uploads, and network scanning activities.

Elastic Agent continuously monitors the custom log file and forwards the collected logs to Elasticsearch. Before indexing, the logs pass through an ingest pipeline where a Grok processor extracts important fields such as timestamp, operator, action, result, device ID, application, and area.

After the logs are indexed into Elasticsearch, Kibana is used to visualize the data through dashboards and Lens visualizations. The Elastic Security module analyzes the indexed events using custom detection rules to generate alerts for suspicious activities. This workflow demonstrates the complete lifecycle of log collection, processing, analysis, visualization, and threat detection within a simplified Security Operations Center.

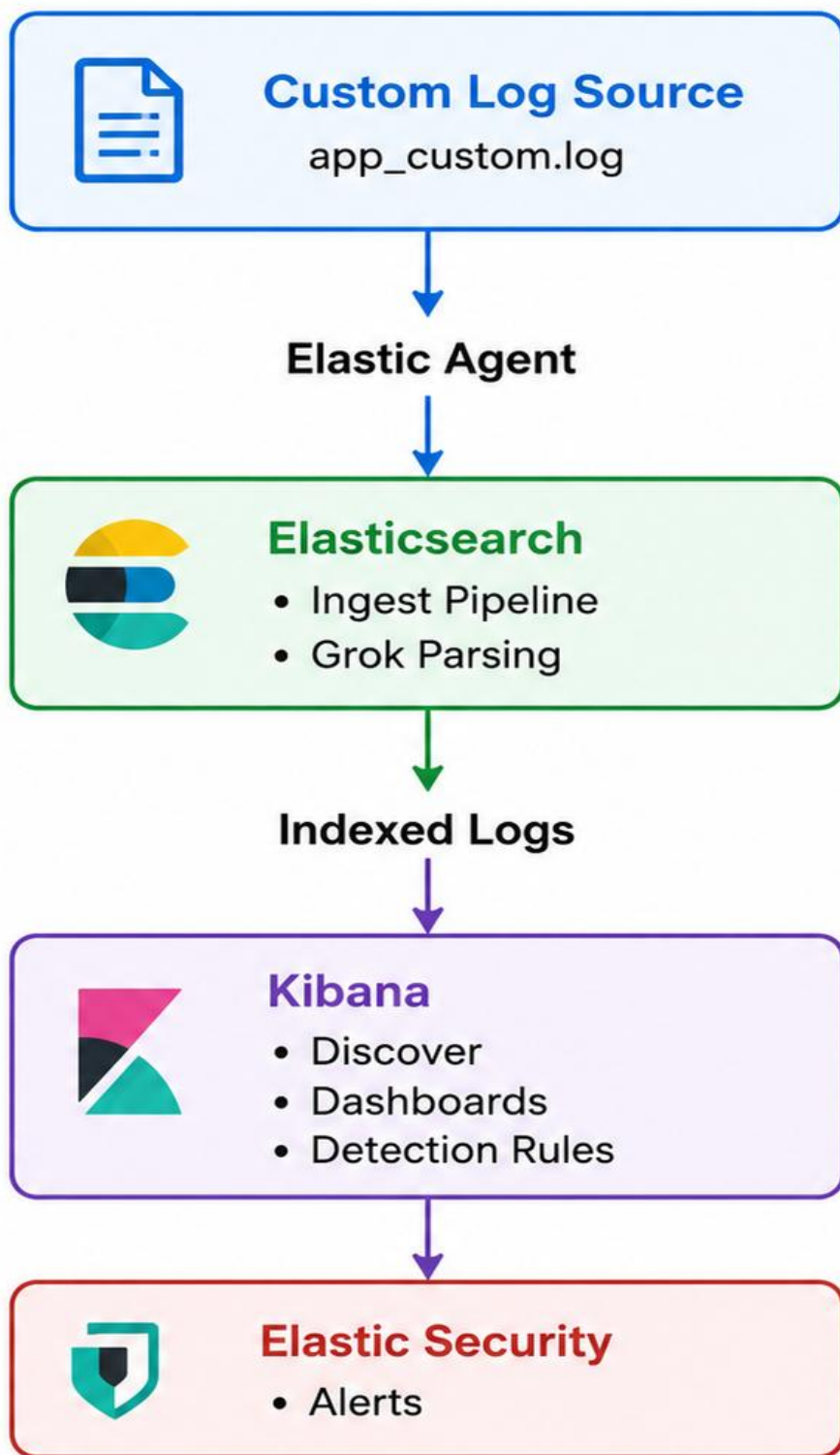


Figure 3.1 – Mini SOC Project Architecture

Figure 3.1 illustrates the architecture of the Mini SOC project. Custom application logs generated by the application are collected by Elastic Agent and forwarded to Elasticsearch. The ingest pipeline uses Grok parsing to extract structured fields before indexing the logs. Kibana provides visualization, dashboards, and detection rules, while Elastic Security generates alerts for suspicious activities.

CHAPTER 4

PROJECT IMPLEMENTATION

4.1 Environment Setup

The Mini SOC environment was deployed using Oracle VirtualBox. Ubuntu Server 24.04 LTS was configured as the primary server to host the Elastic Stack components. Elasticsearch, Kibana, Fleet Server, and Elastic Agent were installed and configured on the Ubuntu system. Windows was used as the client machine for accessing the Kibana web interface and managing the project documentation.

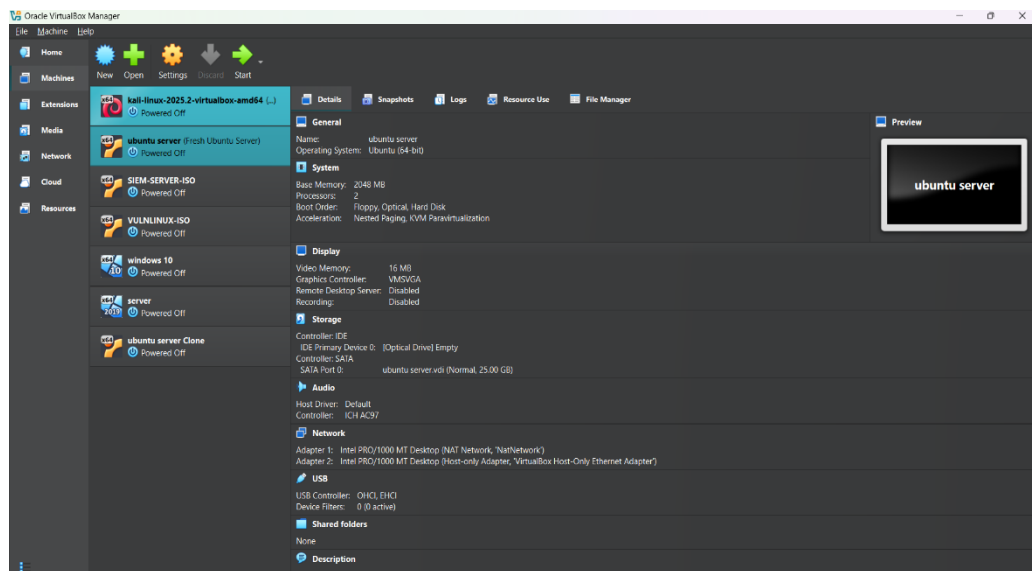
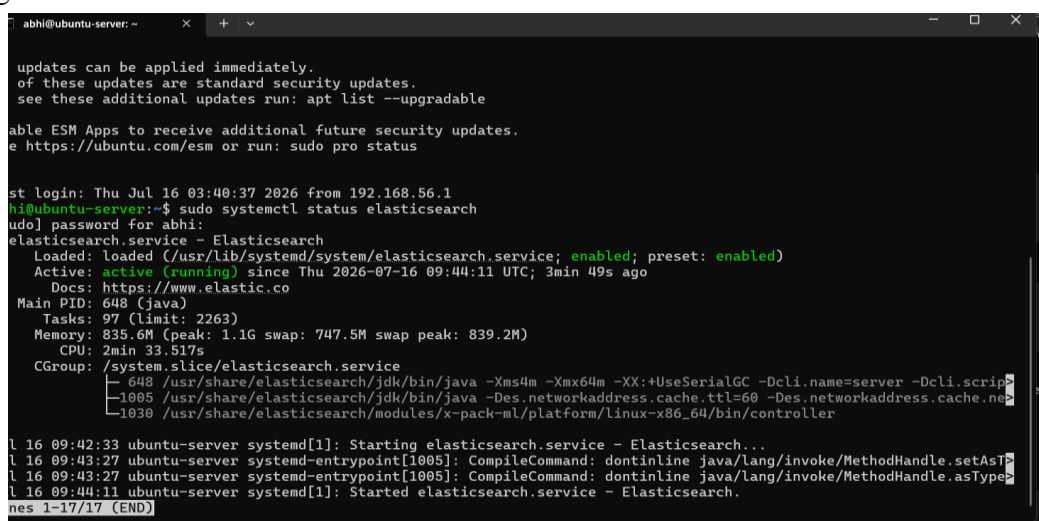


Figure 4.1 – Lab Environment

4.2 Elasticsearch Installation

Elasticsearch was installed on the Ubuntu server to act as the centralized storage and indexing engine for the Mini SOC environment. After installation, the service was configured to start automatically at system boot. Network settings were updated to allow Kibana and other Elastic Stack components to communicate with the Elasticsearch instance. Successful service startup was verified before proceeding with the remaining Elastic Stack configuration.

A terminal window titled 'abhi@ubuntu-server: ~' showing the command 'sudo systemctl status elasticsearch' and its output. The output indicates the service is loaded and active (running). It also shows system logs for the service startup, including the command to start the service and the compilation of the Java command. The terminal output is as follows:

```
updates can be applied immediately.
of these updates are standard security updates.
see these additional updates run: apt list --upgradable

able ESM Apps to receive additional future security updates.
e https://ubuntu.com/esm or run: sudo pro status

st login: Thu Jul 16 03:40:37 2026 from 192.168.56.1
hi@ubuntu-server:~$ sudo systemctl status elasticsearch
udo] password for abhi:
elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-16 09:44:11 UTC; 3min 49s ago
     Docs: https://www.elastic.co
  Main PID: 648 (java)
    Tasks: 97 (limit: 2263)
  Memory: 835.6M (peak: 1.1G swap: 747.5M swap peak: 839.2M)
     CPU: 2min 33.517s
    CGroup: /system.slice/elasticsearch.service
            └─ 648 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:+UseSerialGC -Dcli.name=server -Dcli.scrip
               1005 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cache.ne
               1030 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/bin/controller

l 16 09:42:33 ubuntu-server systemd[1]: Starting elasticsearch.service - Elasticsearch...
l 16 09:43:27 ubuntu-server systemd-entrypoint[1005]: CompileCommand: dontinline java/lang/invoke/MethodHandle.setAsT
l 16 09:43:27 ubuntu-server systemd-entrypoint[1005]: CompileCommand: dontinline java/lang/invoke/MethodHandle.asType
l 16 09:44:11 ubuntu-server systemd[1]: Started elasticsearch.service - Elasticsearch.
nes 1-17/17 (END)
```

Figure 4.2 – Elasticsearch Service Running

4.3 Kibana Configuration

Kibana was configured as the visualization and management interface of the Elastic Stack. The application was connected to the Elasticsearch instance using the generated enrollment token. After successful configuration, Kibana provided access to Discover, Dashboard, Fleet, and Elastic Security applications, enabling centralized monitoring and analysis of ingested logs.

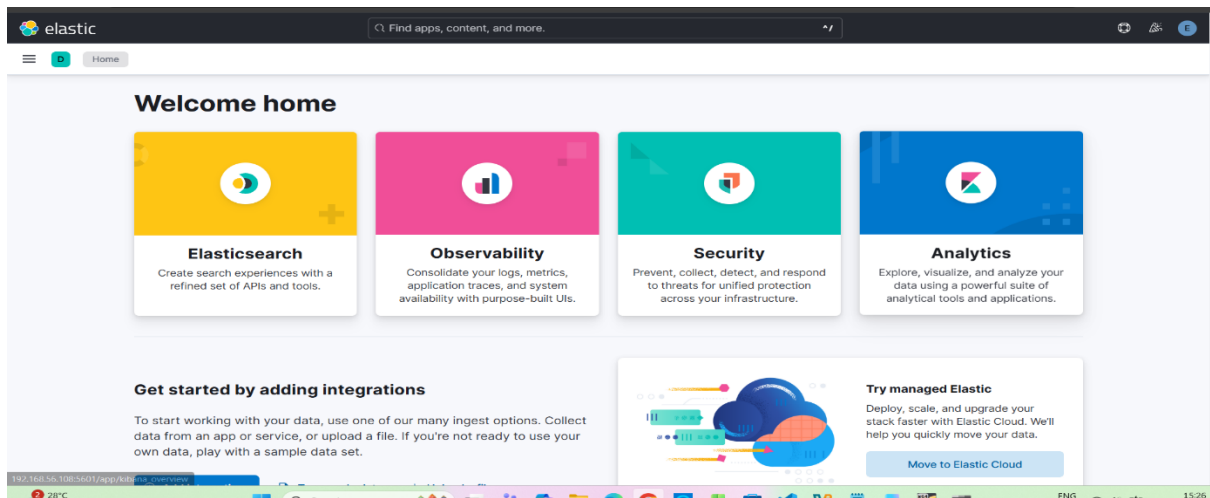
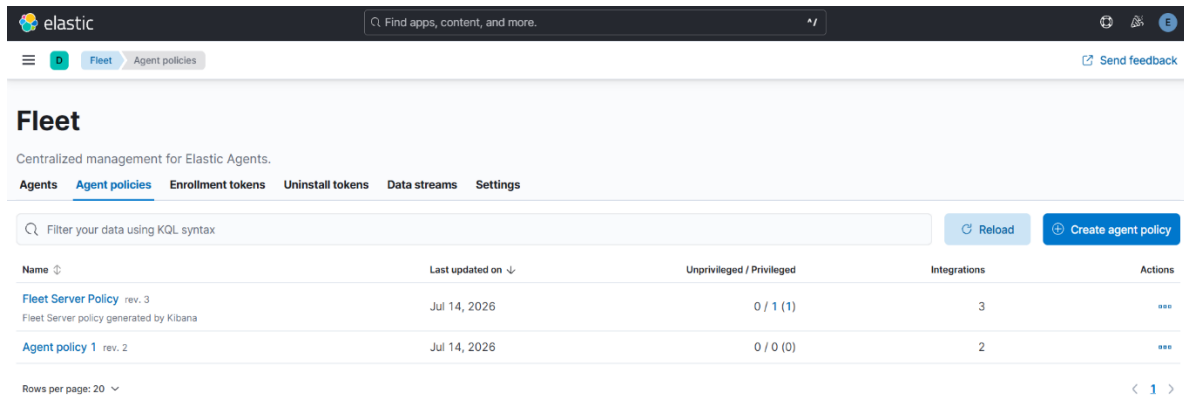


Figure 4.3 – Kibana Configuration

4.4 Fleet Server Configuration

Fleet Server was configured to centrally manage Elastic Agents deployed in the environment. A dedicated Fleet policy was created and assigned to the agent. This enabled centralized management of log collection settings, policy updates, and communication between Elastic Agents and the Elastic Stack.



The screenshot displays the Elastic Fleet Server Agent Policies Configuration page. The page has a dark header with the Elastic logo and a search bar. Below the header, there's a navigation bar with tabs for Agents, Agent policies (selected), Enrollment tokens, Uninstall tokens, Data streams, and Settings. A search bar is present with the placeholder text 'Filter your data using KQL syntax'. To the right of the search bar are buttons for 'Reload' and 'Create agent policy'. The main content area shows a table of agent policies.

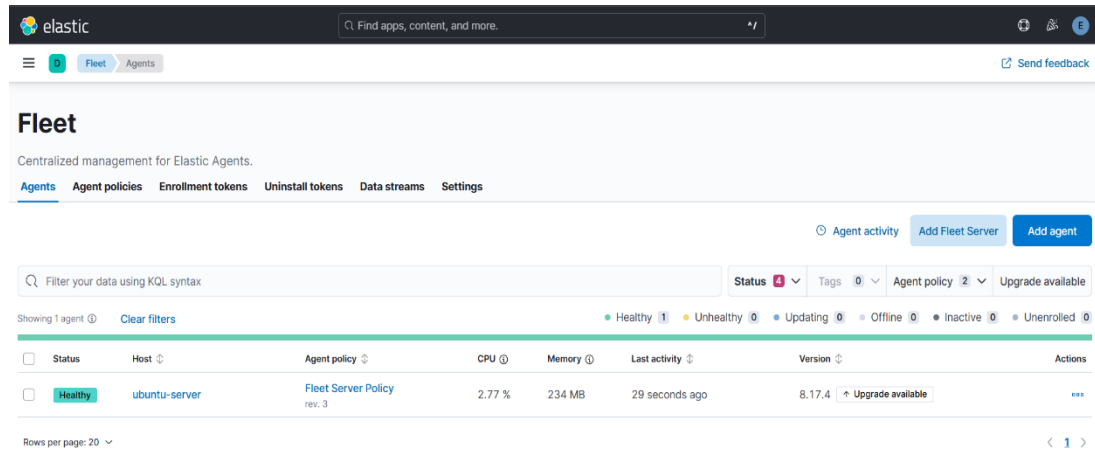
Name	Last updated on	Unprivileged / Privileged	Integrations	Actions
Fleet Server Policy rev. 3 Fleet Server policy generated by Kibana	Jul 14, 2026	0 / 1 (1)	3	...
Agent policy 1 rev. 2	Jul 14, 2026	0 / 0 (0)	2	...

Rows per page: 20

Figure 4.4 – Fleet Server Agent Policies Configuration

4.5 Elastic Agent Installation

Elastic Agent was installed on the Ubuntu server responsible for generating the custom application logs. The agent continuously monitored the specified log file using the Filestream integration and securely forwarded new log entries to Elasticsearch through Fleet Server.



The screenshot displays the Elastic Fleet management console. At the top, the 'elastic' logo and a search bar are visible. Below the navigation bar, the 'Fleet' section is active, showing 'Centralized management for Elastic Agents.' and tabs for 'Agents', 'Agent policies', 'Enrollment tokens', 'Uninstall tokens', 'Data streams', and 'Settings'. The 'Agents' tab is selected, showing a table of enrolled agents. The table has columns for Status, Host, Agent policy, CPU, Memory, Last activity, Version, and Actions. A single agent is listed with the status 'Healthy', host 'ubuntu-server', and policy 'Fleet Server Policy rev. 3'. The version is 8.17.4, and an 'Upgrade available' button is shown. The bottom of the page indicates 'Rows per page: 20' and a pagination control showing page 1 of 1.

Status	Host	Agent policy	CPU	Memory	Last activity	Version	Actions
Healthy	ubuntu-server	Fleet Server Policy rev. 3	2.77 %	234 MB	29 seconds ago	8.17.4 Upgrade available	...

Figure 4.5 – Elastic Agent Successfully Enrolled in Fleet

4.6 Custom Log Generation

Custom application logs were generated to simulate security-related events in a controlled environment. Each log entry follows a structured format containing important information required for monitoring and threat detection. The generated logs include both normal and suspicious activities such as successful logins, failed login attempts, firmware uploads, network scans, and USB device detection.

Timestamp – Records the exact date and time of the event.

Host – Identifies the system where the event occurred.

Application – Specifies the application generating the log.

Severity – Indicates the importance of the event (INFO, WARNING, ERROR).

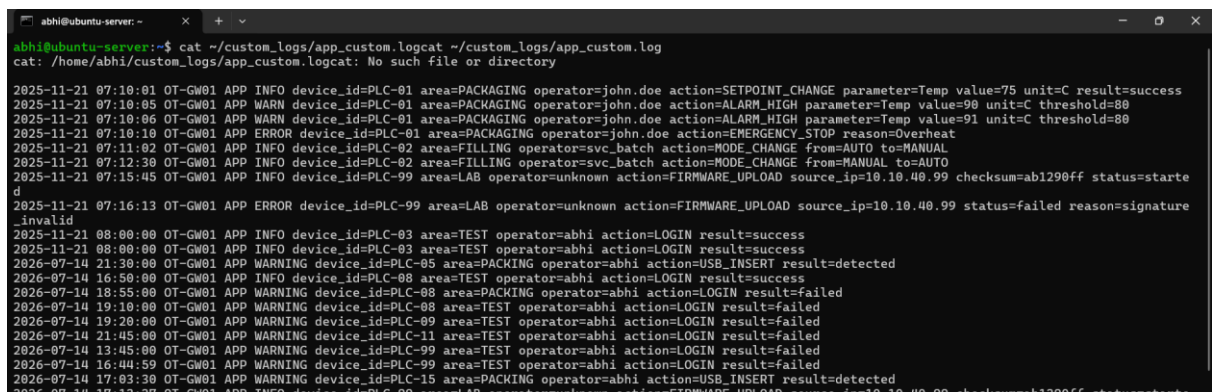
Device ID – Identifies the PLC or device involved in the event.

Area – Represents the operational section where the activity occurred.

Operator – Identifies the user performing the action.

Action – Records the operation performed.

Result – Indicates whether the action was successful or failed.



```
abhi@ubuntu-server:~$ cat ~/custom_logs/app_custom.logcat ~/custom_logs/app_custom.log
cat: /home/abhi/custom_logs/app_custom.logcat: No such file or directory

2025-11-21 07:10:01 OT-GW01 APP INFO device_id=PLC-01 area=PACKAGING operator=john.doe action=SETPOINT_CHANGE parameter=Temp value=75 unit=C result=success
2025-11-21 07:10:05 OT-GW01 APP WARN device_id=PLC-01 area=PACKAGING operator=john.doe action=ALARM_HIGH parameter=Temp value=90 unit=C threshold=80
2025-11-21 07:10:06 OT-GW01 APP WARN device_id=PLC-01 area=PACKAGING operator=john.doe action=ALARM_HIGH parameter=Temp value=91 unit=C threshold=80
2025-11-21 07:10:10 OT-GW01 APP ERROR device_id=PLC-01 area=PACKAGING operator=john.doe action=EMERGENCY_STOP reason=Overheat
2025-11-21 07:11:02 OT-GW01 APP INFO device_id=PLC-02 area=FILLING operator=svc_batch action=MODE_CHANGE from=AUTO to=MANUAL
2025-11-21 07:12:30 OT-GW01 APP INFO device_id=PLC-02 area=FILLING operator=svc_batch action=MODE_CHANGE from=MANUAL to=AUTO
2025-11-21 07:15:45 OT-GW01 APP INFO device_id=PLC-99 area=LAB operator=unknown action=FIRMWARE_UPLOAD source_ip=10.10.40.99 checksum=ab1290ff status=started
2025-11-21 07:16:13 OT-GW01 APP ERROR device_id=PLC-99 area=LAB operator=unknown action=FIRMWARE_UPLOAD source_ip=10.10.40.99 status=failed reason=signature_invalid
2025-11-21 08:00:00 OT-GW01 APP INFO device_id=PLC-03 area=TEST operator=abhi action=LOGIN result=success
2025-11-21 08:00:00 OT-GW01 APP INFO device_id=PLC-03 area=TEST operator=abhi action=LOGIN result=success
2026-07-14 21:30:00 OT-GW01 APP WARNING device_id=PLC-05 area=PACKING operator=abhi action=USB_INSERT result=detected
2026-07-14 16:50:00 OT-GW01 APP INFO device_id=PLC-08 area=TEST operator=abhi action=LOGIN result=success
2026-07-14 18:55:00 OT-GW01 APP WARNING device_id=PLC-08 area=PACKING operator=abhi action=LOGIN result=failed
2026-07-14 19:10:00 OT-GW01 APP WARNING device_id=PLC-08 area=TEST operator=abhi action=LOGIN result=failed
2026-07-14 19:20:00 OT-GW01 APP WARNING device_id=PLC-09 area=TEST operator=abhi action=LOGIN result=failed
2026-07-14 21:45:00 OT-GW01 APP WARNING device_id=PLC-11 area=TEST operator=abhi action=LOGIN result=failed
2026-07-14 13:45:00 OT-GW01 APP WARNING device_id=PLC-99 area=TEST operator=abhi action=LOGIN result=failed
2026-07-14 16:44:59 OT-GW01 APP WARNING device_id=PLC-99 area=TEST operator=abhi action=LOGIN result=failed
2026-07-14 17:03:30 OT-GW01 APP WARNING device_id=PLC-15 area=PACKING operator=abhi action=USB_INSERT result=detected
2026-07-14 17:13:27 OT-GW01 APP INFO device_id=PLC-99 area=LAB operator=unknown action=FIRMWARE_UPLOAD source_ip=10.10.40.99 checksum=ab1290ff status=started
```

Figure 4.6 – Generated Custom Application Logs

4.7 Ingest Pipeline and Grok Parsing

An Elasticsearch ingest pipeline was configured using the Grok processor to automatically parse the raw application logs before indexing. The pipeline extracts important information from each log entry and stores it as individual fields, eliminating the need to manually search through raw log messages.

The Grok processor separates the log into structured fields such as timestamp, host name, application, severity, device ID, area, operator, action, and result. These extracted fields are stored following the Elastic Common Schema (ECS) where applicable, improving consistency across the Elastic Stack.

After parsing, the indexed fields become searchable and can be used for filtering, dashboard visualizations, and detection rules in Kibana.

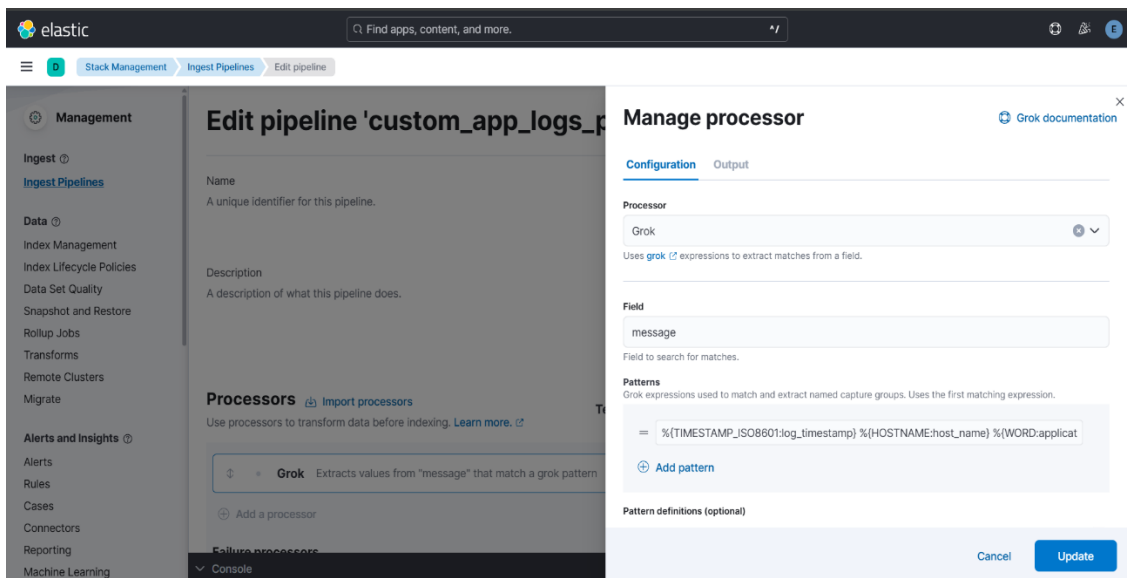


Figure 4.7 – Ingest Pipeline with Grok Processor Configuration

4.8 Data Verification in Discover

The Kibana Discover application was used to verify that the custom application logs were successfully collected, parsed, and indexed in Elasticsearch. Individual fields extracted by the Grok processor were examined to ensure that the parsing process correctly separated the log data into structured values.

The Discover interface was also used to perform searches and apply filters based on fields such as action, result, operator, device ID, and area. Successful verification confirmed that the ingest pipeline was functioning correctly and that the indexed data was ready for dashboard creation and automated threat detection.

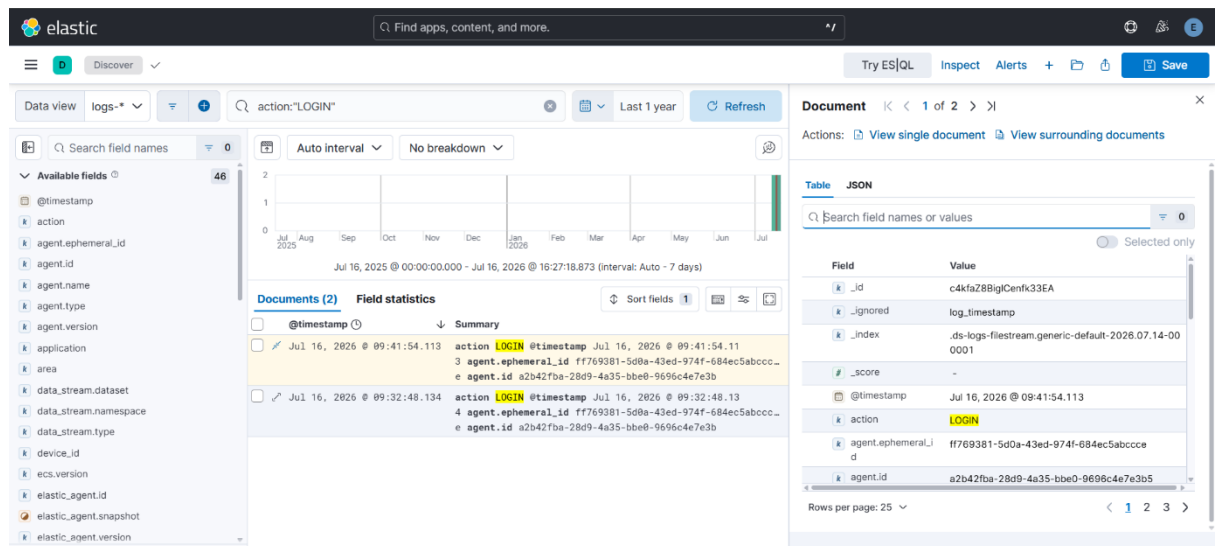


Figure 4.8 – Parsed Log Data Verified in Kibana Discover

CHAPTER 5

DASHBOARD DEVELOPMENT

5.1 Mini SOC Dashboard Overview

The Mini SOC Dashboard provides a centralized view of security events collected from the custom application logs. It combines multiple visualizations, including login activity trends, failed login attempts, total security events, source IP analysis, and operator-based login activity, allowing analysts to monitor and investigate events from a single interface.

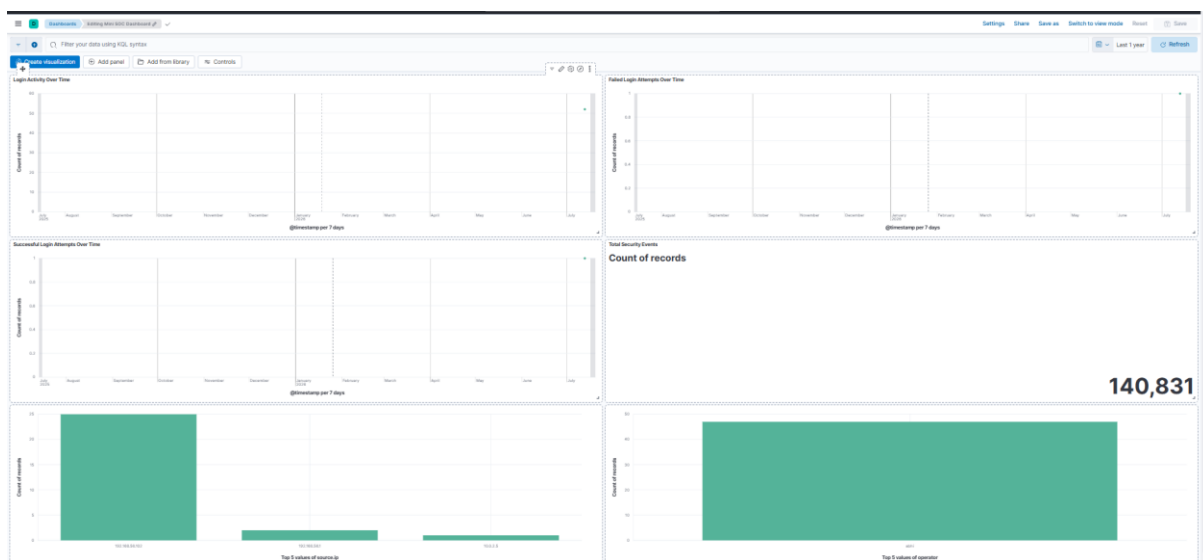


Figure 5.1 – Mini SOC Dashboard Overview

5.2 Login Activity Over Time

A line chart visualization was created to display login activity over time. This visualization helps identify authentication trends and unusual spikes in login events.

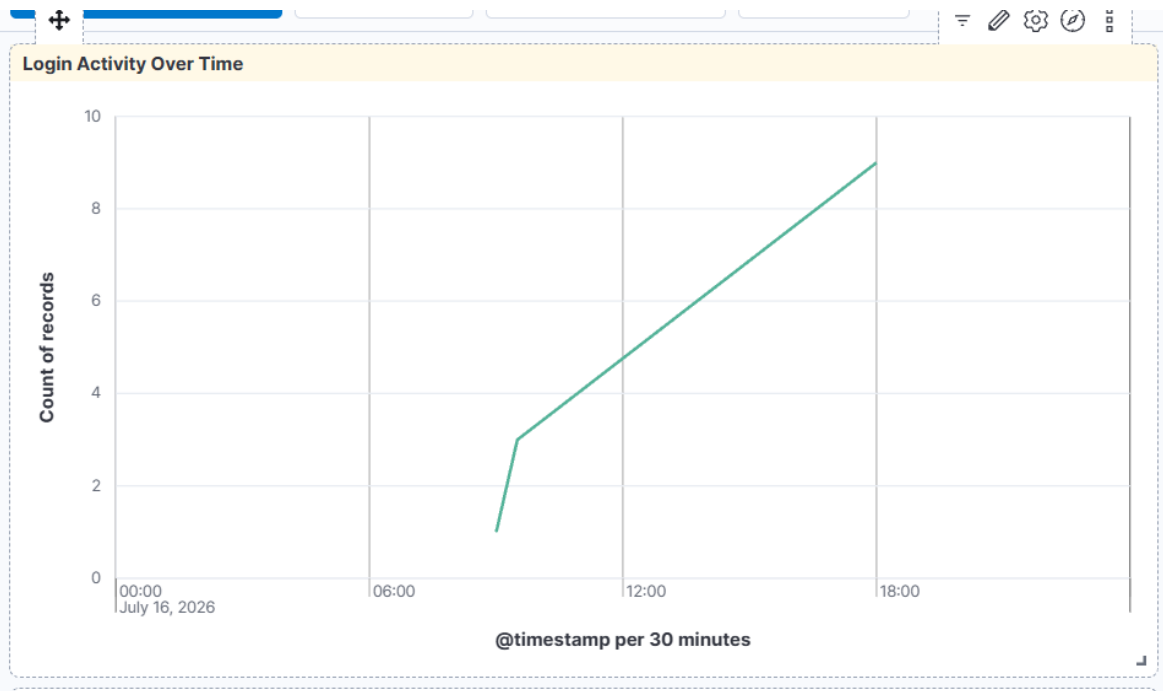


Figure 5.2 – Login Activity Over Time

5.3 Failed Login Attempts Over Time

A line chart visualization was created to display failed login attempts over time. This visualization helps analysts identify failed authentication activity and detect possible unauthorized access attempts.

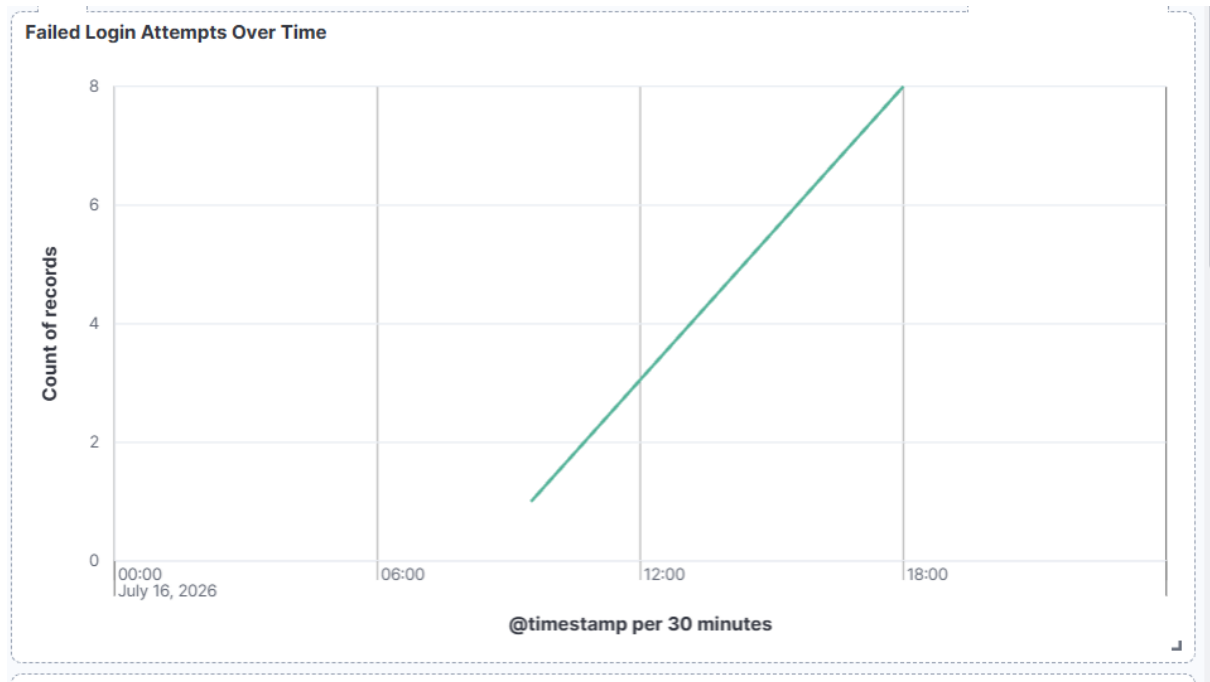


Figure 5.3 – Failed Login Attempts Over Time

5.4 Total Security Events

A metric visualization was created to display the total number of security events collected from the custom application logs. This provides a quick overview of the overall event volume within the monitored environment.

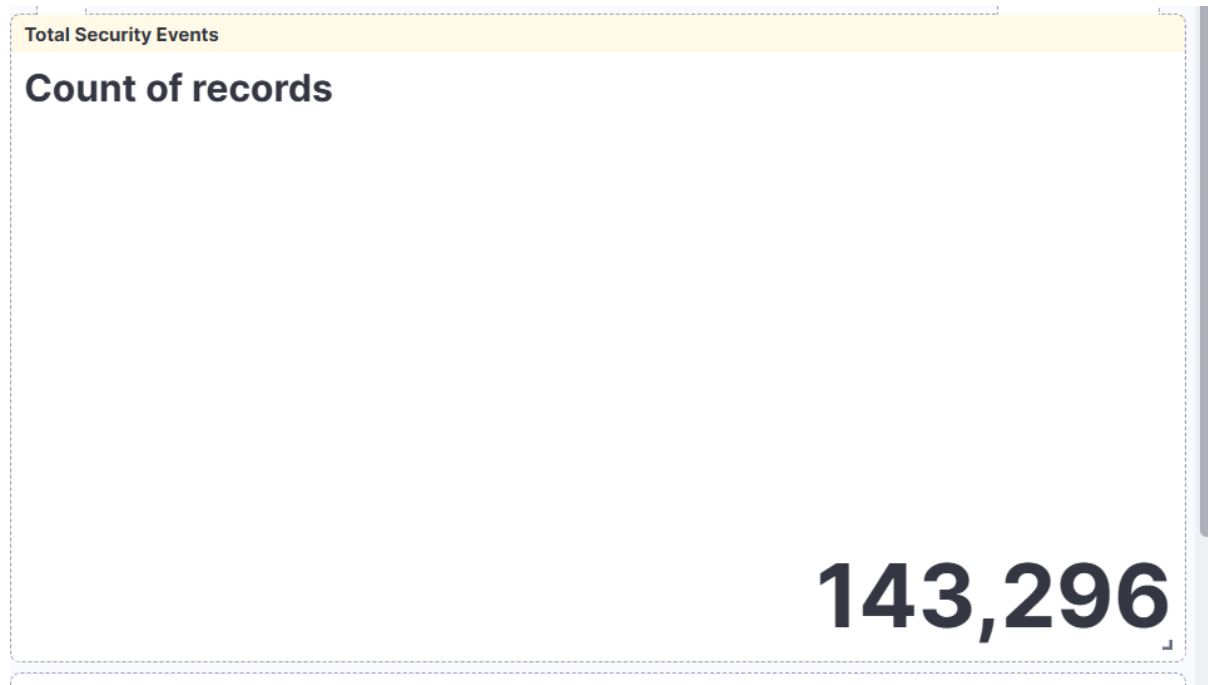


Figure 5.4 – Total Security Events

5.5 Top Source IP Addresses

A bar chart visualization was created to display the top source IP addresses generating security events. This helps analysts identify the most active systems communicating with the monitored application.

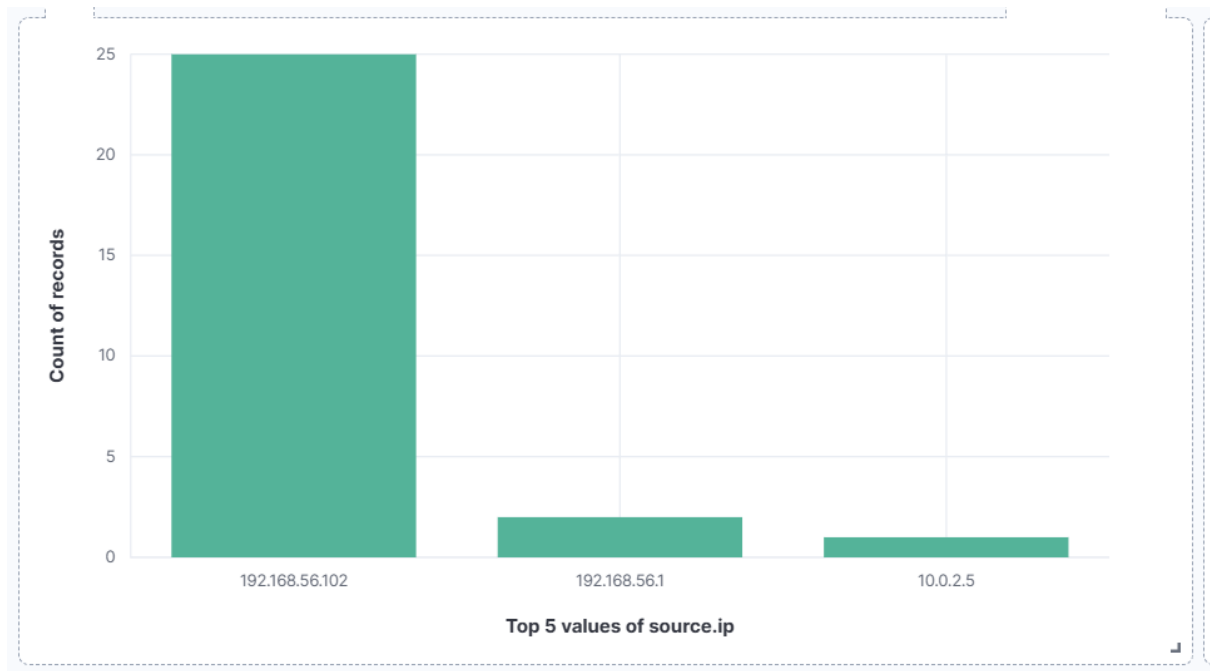


Figure 5.5 – Top Source IP Addresses

5.6 Login Attempts by Operator

A bar chart visualization was created to display login attempts grouped by operator. This helps identify which users generate the highest number of authentication events and assists in detecting unusual user activity.

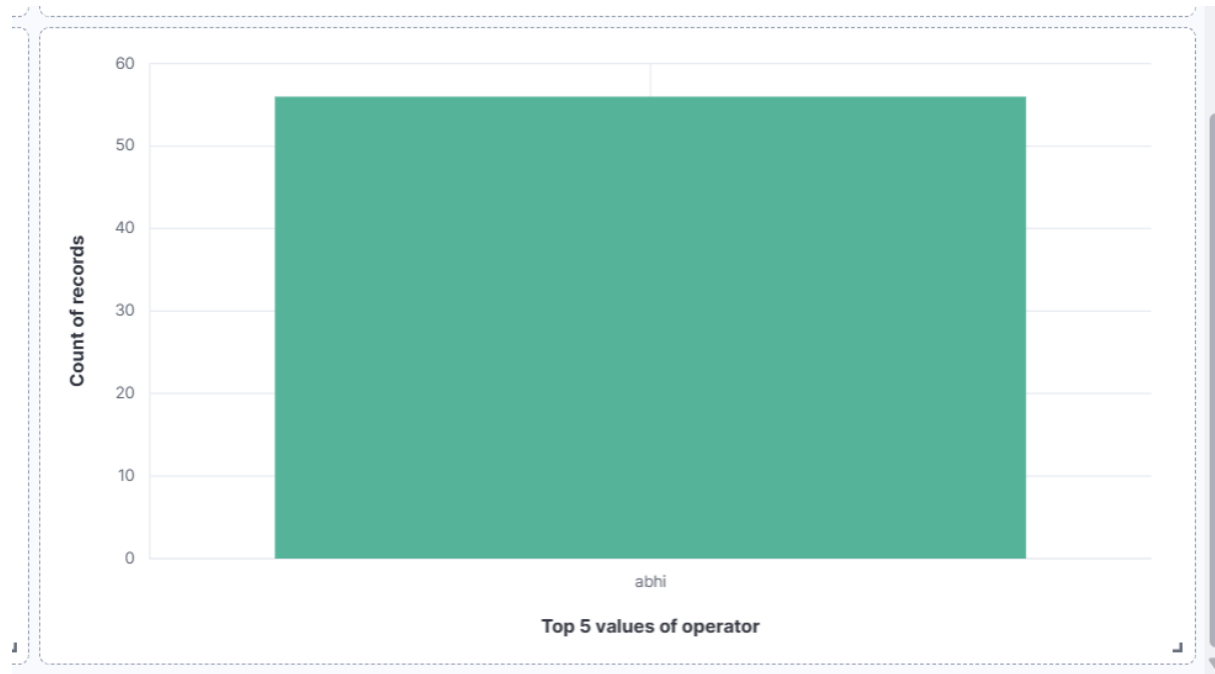


Figure 5.6 – Login Attempts by Operator

5.7 Compliance Dashboard Overview

A Compliance Dashboard was developed in Kibana to provide a centralized view of authentication events generated by the custom application logs. The dashboard includes visualizations for total login attempts, successful and failed logins, login trends, device activity, and login status distribution, enabling efficient monitoring and audit of user authentication activities.

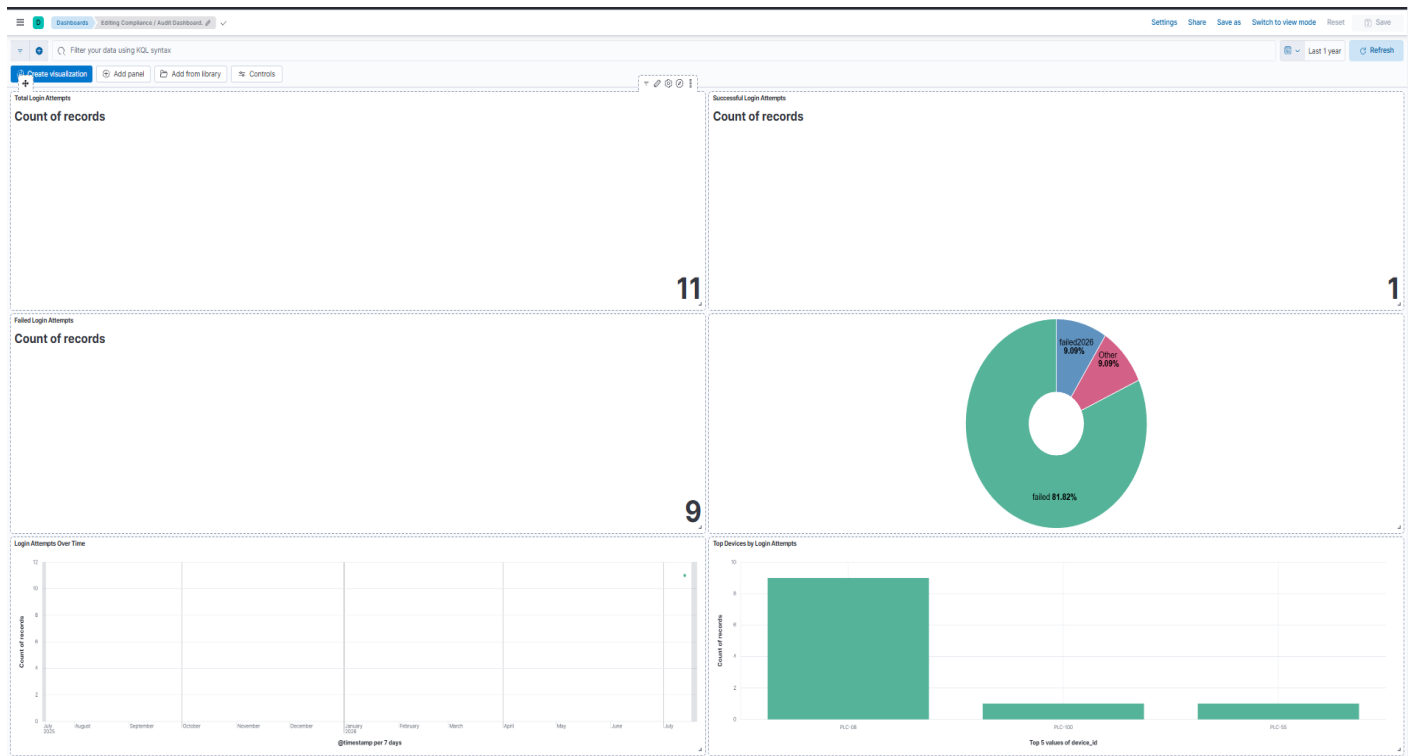


Figure 5.7 – Compliance Dashboard Overview

5.8 Total Login Attempts

A metric visualization was created to display the total number of login attempts recorded in the collected logs. This provides a quick overview of authentication activity within the monitored environment.

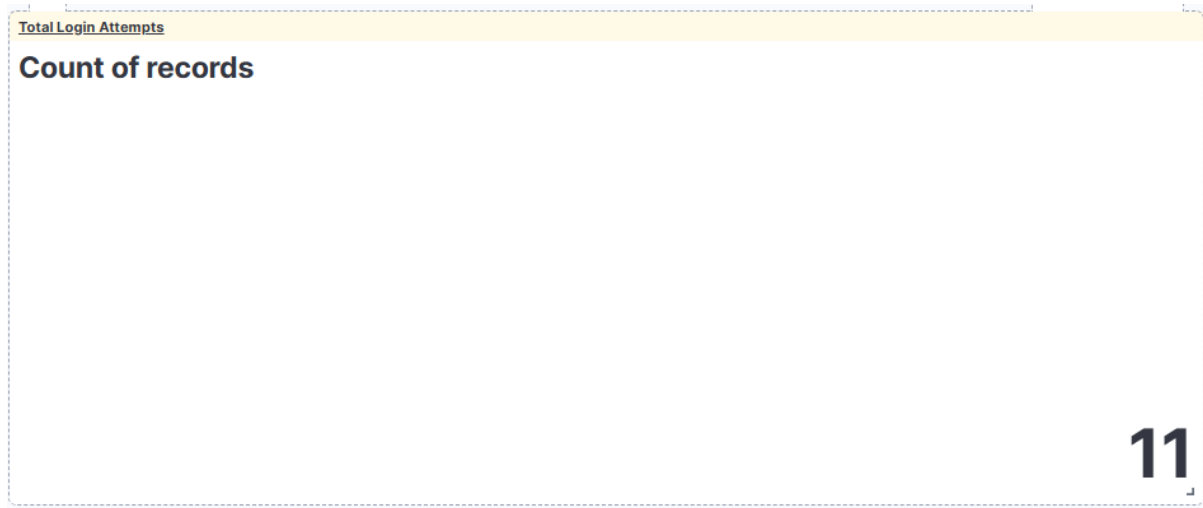


Figure 5.8 – Total Login Attempts

5.9 Successful Login Attempts

A metric visualization was created to display the total number of successful login attempts. This enables analysts to monitor successful user authentications and compare them with failed login events.

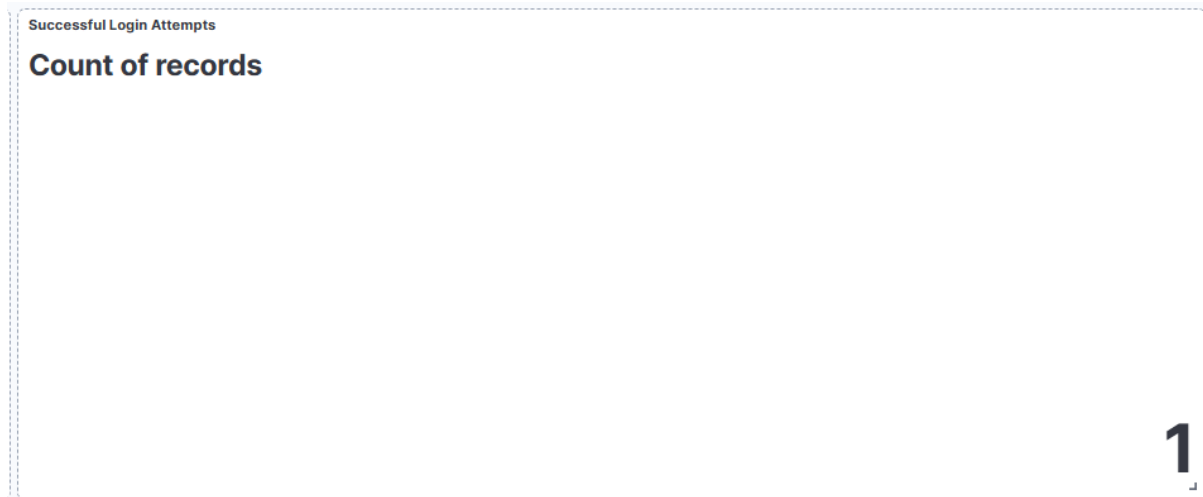


Figure 5.9 – Successful Login Attempts

5.10 Failed Login Attempts

A metric visualization was created to display the total number of failed login attempts. This helps identify authentication failures that may indicate incorrect credentials or unauthorized access attempts.

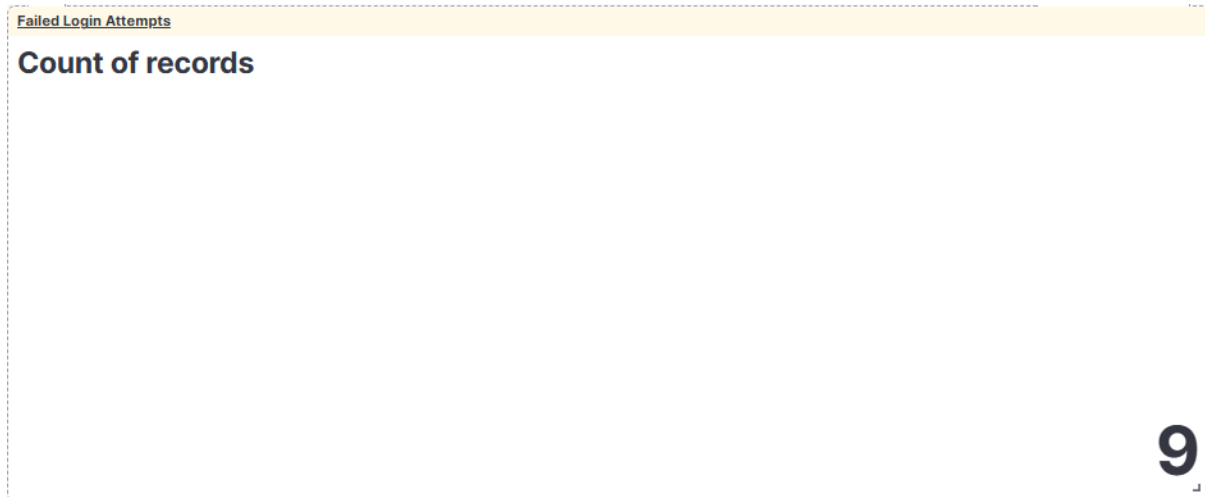


Figure 5.10 – Failed Login Attempts

5.11 Successful vs Failed Login Attempts

A pie chart visualization was created to compare successful and failed login attempts. This visualization provides an overview of authentication outcomes and helps identify unusual login failure patterns.

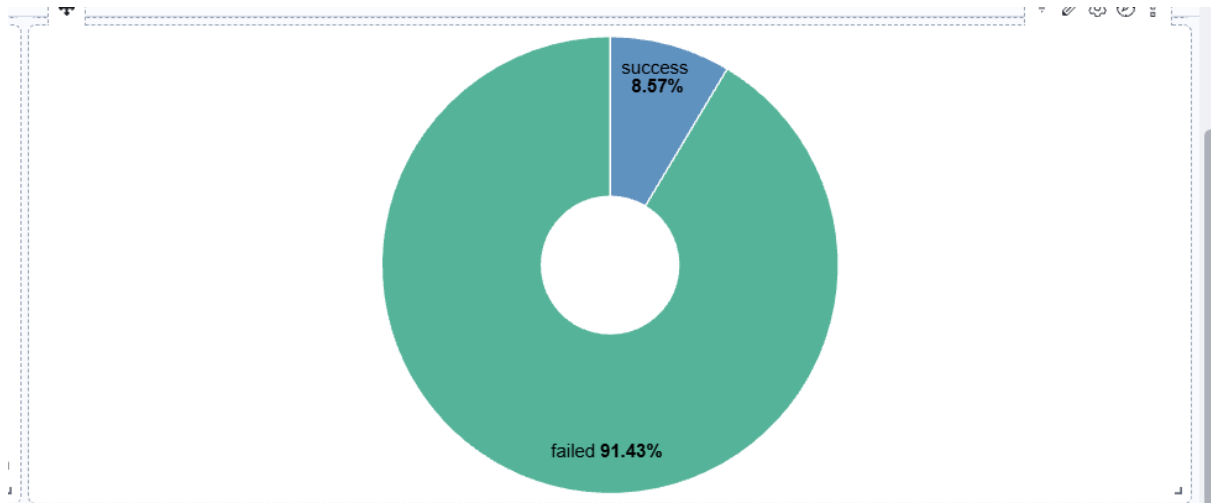


Figure 5.11 – Successful vs Failed Login Attempts

5.12 Top Devices by Login Attempts

A bar chart visualization was created to display the devices generating the highest number of login attempts. This visualization helps analysts identify the most active devices, detect unusual authentication activity, and monitor device-based login patterns within the monitored environment.

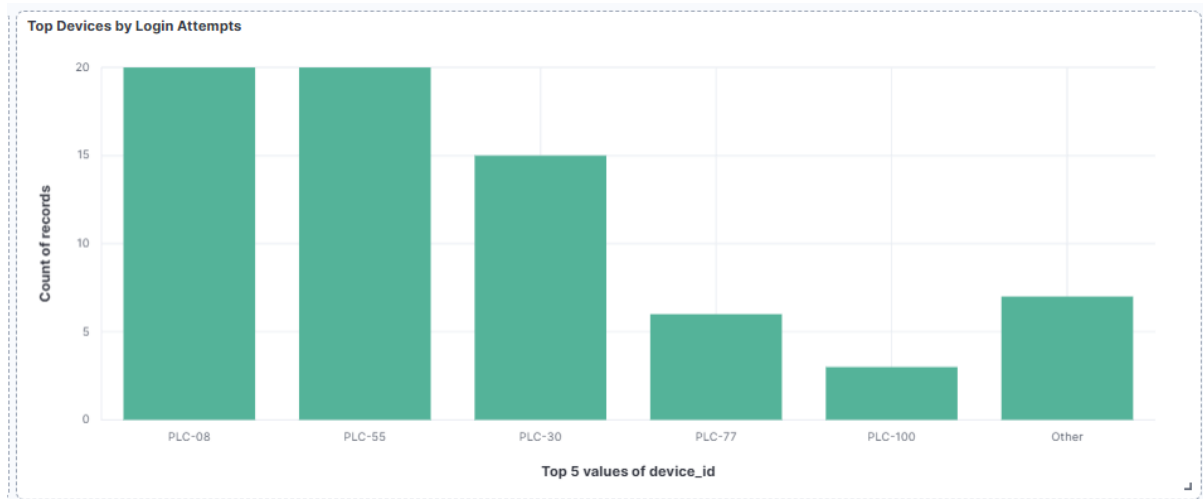


Figure 5.12 – Top Devices by Login Attempts

CHAPTER 6

DETECTION RULES

6.1 Detection Rule Overview

Detection rules were created in the Elastic Security module to automatically identify suspicious activities from the collected logs. These rules continuously monitor indexed events and generate alerts whenever predefined conditions are met. The generated alerts assist security analysts in identifying potential threats and responding to security incidents efficiently.

6.2 Failed Login Detection

A detection rule was created to identify failed login attempts from the custom application logs. The rule monitors events where the login action results in failure and generates an alert whenever such activity is detected. This helps identify unauthorized access attempts and authentication failures.

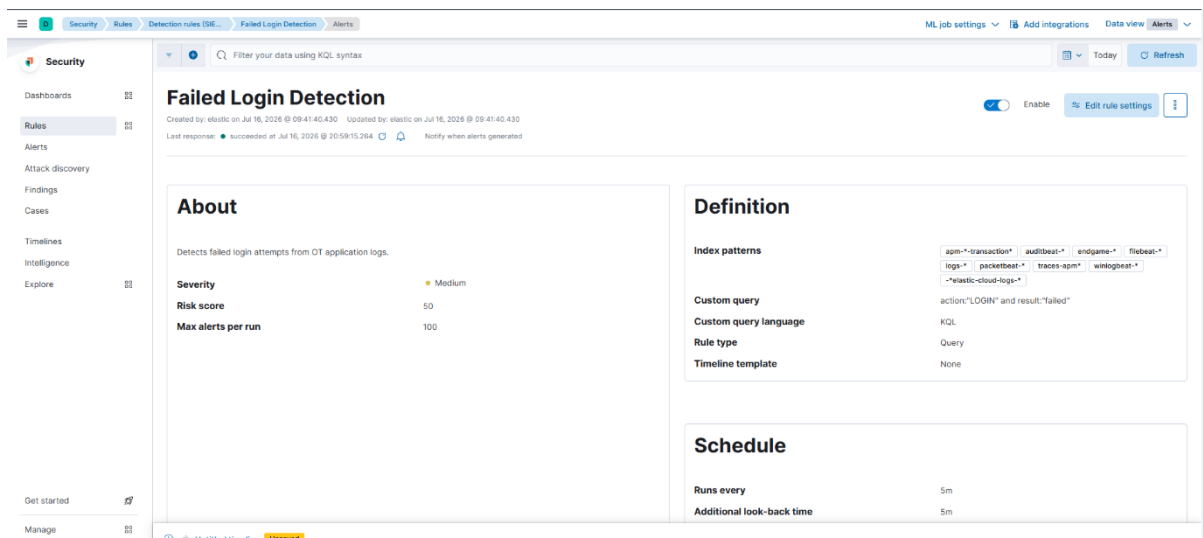


Figure 6.2 – Failed Login Detection Rule

6.3 Multiple Failed Login Detection

A threshold detection rule was configured to identify multiple failed login attempts occurring within a defined time period. This rule helps detect possible brute-force attacks by generating an alert when repeated authentication failures exceed the specified threshold.

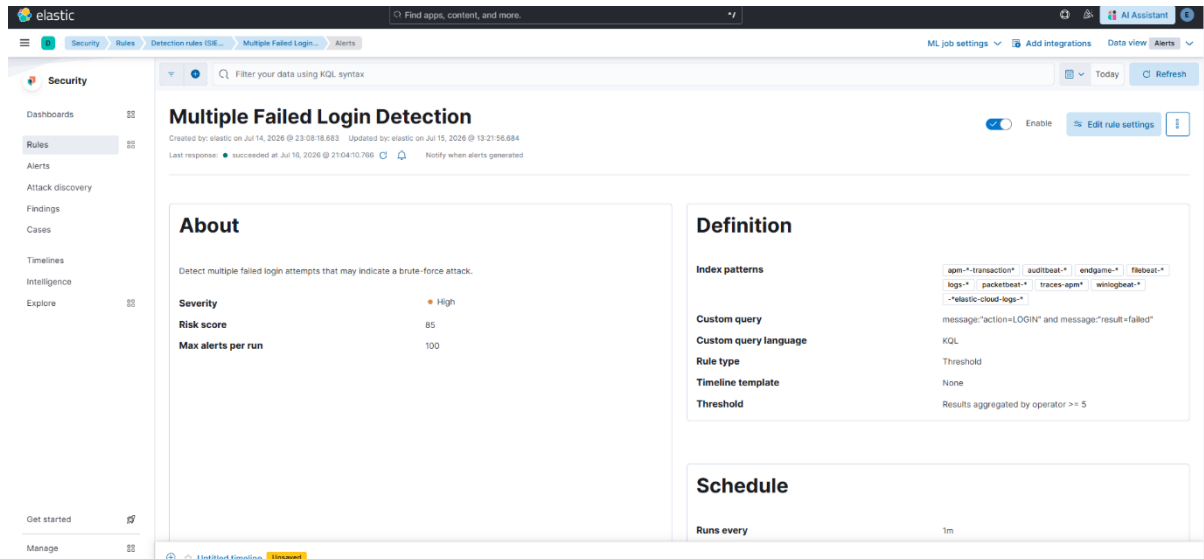


Figure 6.3 – Multiple Failed Login Detection Rule

6.4 Firmware Upload Detection

A custom detection rule was created to monitor firmware upload activities recorded in the application logs. Since firmware updates are sensitive administrative operations, alerts are generated whenever a firmware upload event is detected, allowing analysts to verify whether the activity is authorized.

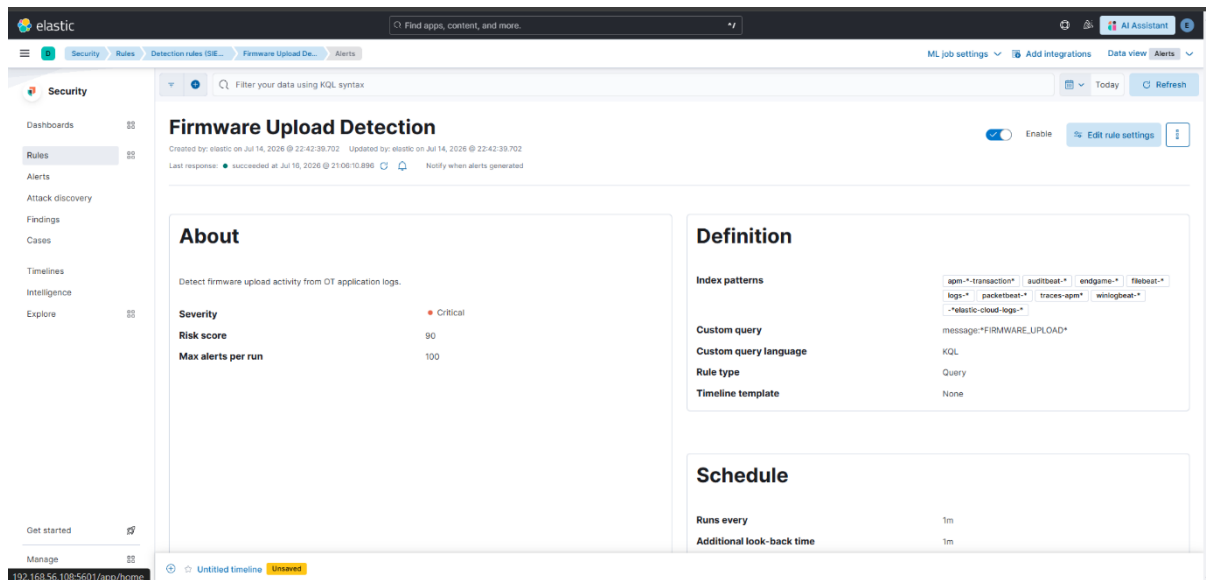


Figure 6.4– Firmware Upload Detection Rule

6.5 Network Scan Detection

A detection rule was implemented to identify network scanning activities performed within the monitored environment. The rule detects events associated with network scan operations and generates alerts to indicate possible reconnaissance attempts before an attack.

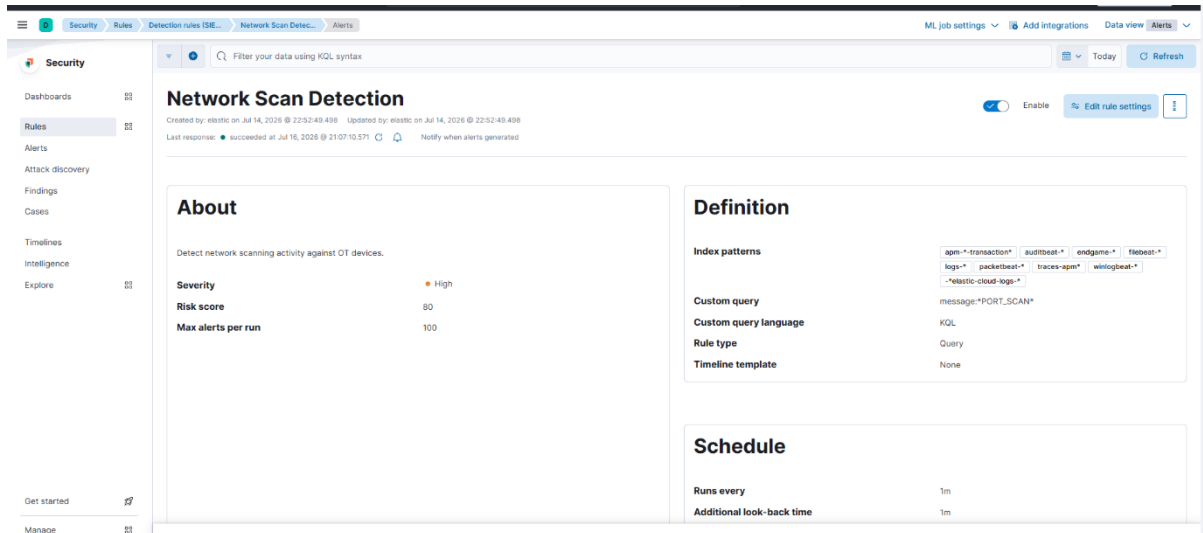


Figure 6.5 – Network Scan Detection Rule

6.6 USB Device Detection

A custom detection rule was created to identify USB device insertion events collected from the monitored environment. The rule monitors USB-related log events and generates an alert whenever a new USB device is detected. This helps analysts identify unauthorized removable media usage and supports monitoring for potential data exfiltration or unauthorized device access.

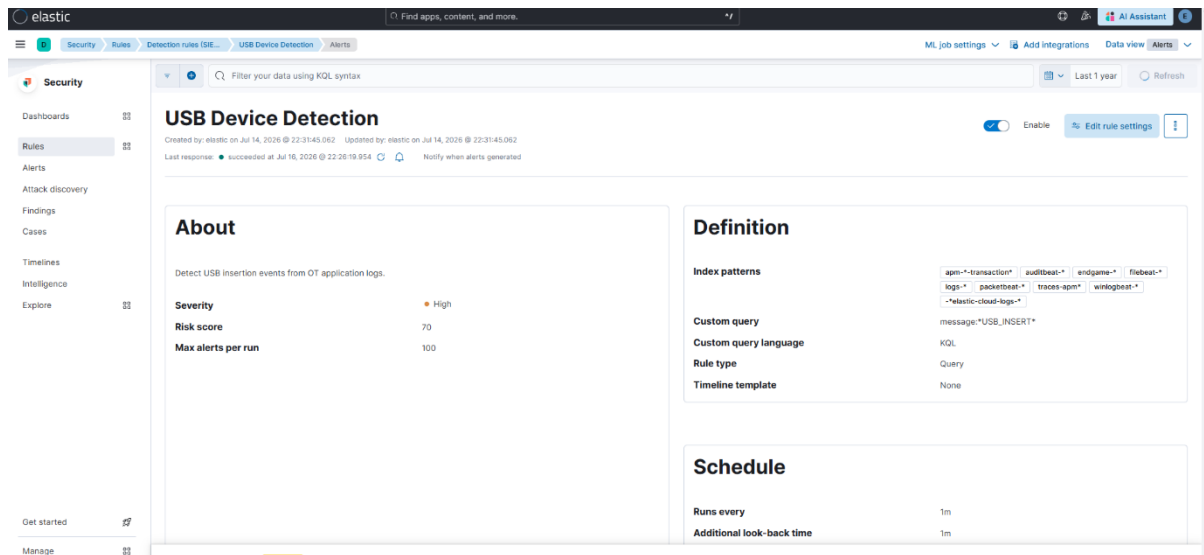


Figure 6.6 – USB Device Detection Rule

6.7 Alert Verification

The generated alerts were verified using the Elastic Security Alerts dashboard. The dashboard confirmed that the configured detection rules successfully generated alerts for suspicious activities, including failed login attempts, firmware upload events, multiple failed login attempts, and USB device detection. This verification demonstrated the successful workflow from log collection and parsing to automated threat detection and alert generation.

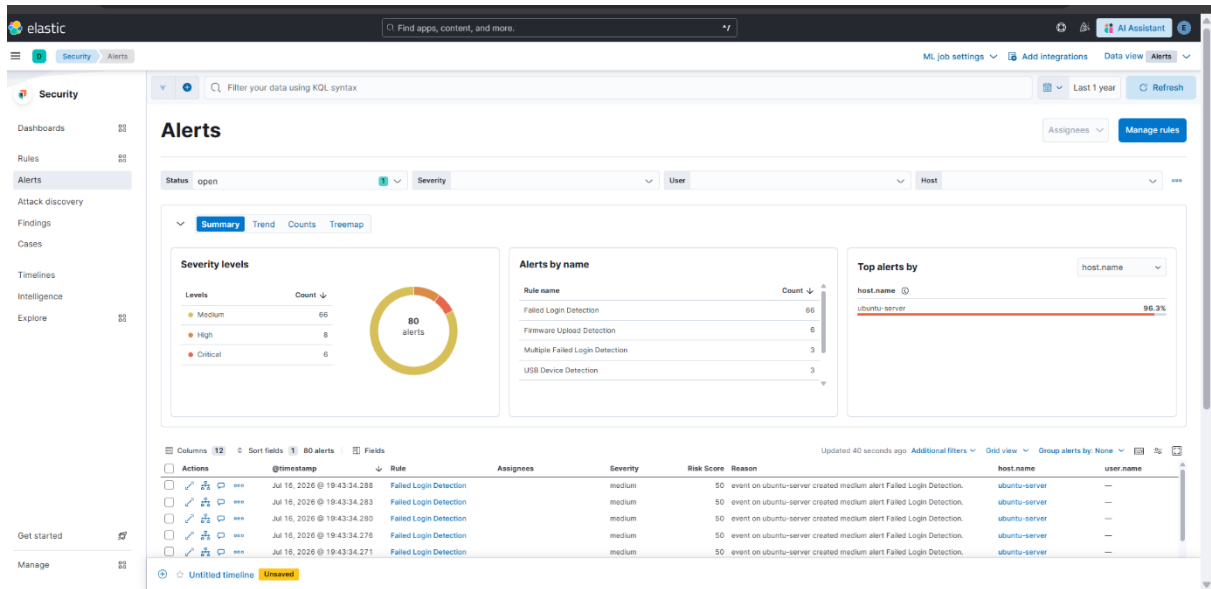


Figure 6.7 – Detection Alerts Verified in Elastic Security

CHAPTER 7

CONCLUSION

This project successfully demonstrated the implementation of a simplified Security Operations Center (SOC) using the Elastic Stack (ELK). The complete workflow of log collection, parsing, indexing, visualization, and threat detection was implemented within a controlled lab environment.

Custom application logs were collected using Elastic Agent and processed through an Ingest Pipeline with Grok patterns to extract structured fields. The processed logs were indexed into Elasticsearch and visualized in Kibana using interactive dashboards that displayed login statistics, authentication trends, and device-based activities.

Custom detection rules were created in Elastic Security to identify failed login attempts, multiple failed logins, firmware upload events, network scanning, and USB device insertion. These rules successfully generated alerts, which were verified through the Elastic Security Alerts dashboard, confirming the effectiveness of the implemented detection workflow.

Overall, this project provided practical experience in SIEM deployment, centralized log management, dashboard creation, and detection engineering using the Elastic Stack. It strengthened my understanding of SOC operations and security monitoring while providing hands-on experience that is valuable for entry-level cybersecurity and SOC Analyst roles.

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